

# Experiment 4: Fire Tolerant Plants Analysis

## Fire-Related Plant Traits Analysis

Total Species with Fire Traits: 1447

Fire-Related Traits Analyzed: 14

*Generated by Fire Traits Analysis Script*

# Fire Risk Assessment Methodology

## FIRE RISK ASSESSMENT METHODOLOGY

### Risk Assessment Criteria:

The fire risk assessment is based on two key fire survival traits:

#### 1. RESPROUTING CAPACITY

- Ability to regrow from surviving plant parts after fire
- Indicates rapid post-fire recovery potential
- Species with 'resprouts' in resprouting\_capacity trait

#### 2. POST-FIRE RECRUITMENT

- Ability to establish new individuals after fire events
- Indicates long-term population recovery potential
- Species with 'post\_fire\_recruitment' (not 'absent') in post\_fire\_recruitment trait

### RISK LEVEL DETERMINATION:

#### LOW RISK (Green):

- Condition: Can BOTH resprout AND has post-fire recruitment
- Interpretation: Best fire survival strategy with dual protection
- Management: Can withstand frequent burning, suitable for hazard reduction burns

#### MEDIUM RISK (Orange):

- Condition: Can EITHER resprout OR has post-fire recruitment (only one trait)
- Interpretation: Partial fire survival strategy with limited protection
- Management: Moderate fire tolerance, requires careful burn planning

#### HIGH RISK (Red):

- Condition: Cannot resprout AND has no post-fire recruitment
- Interpretation: Poor fire survival strategy with minimal protection
- Management: High fire sensitivity, needs protection from frequent burning

### APPLICATION FOR HAZARD REDUCTION BURNING:

- Low Risk species/families: Suitable for frequent burning
- Medium Risk species/families: Moderate burning frequency
- High Risk species/families: Minimal burning, focus on protection

# Fire-Related Plant Species Analysis

